

CHAPTER 3

MATLAB BASICS

WHAT WILL WE LEARN?

- What is MATLAB?
- What programming environment does MATLAB offer?
- What are M-files?
- What is the difference between MATLAB scripts and functions?
- How can I get started with MATLAB?

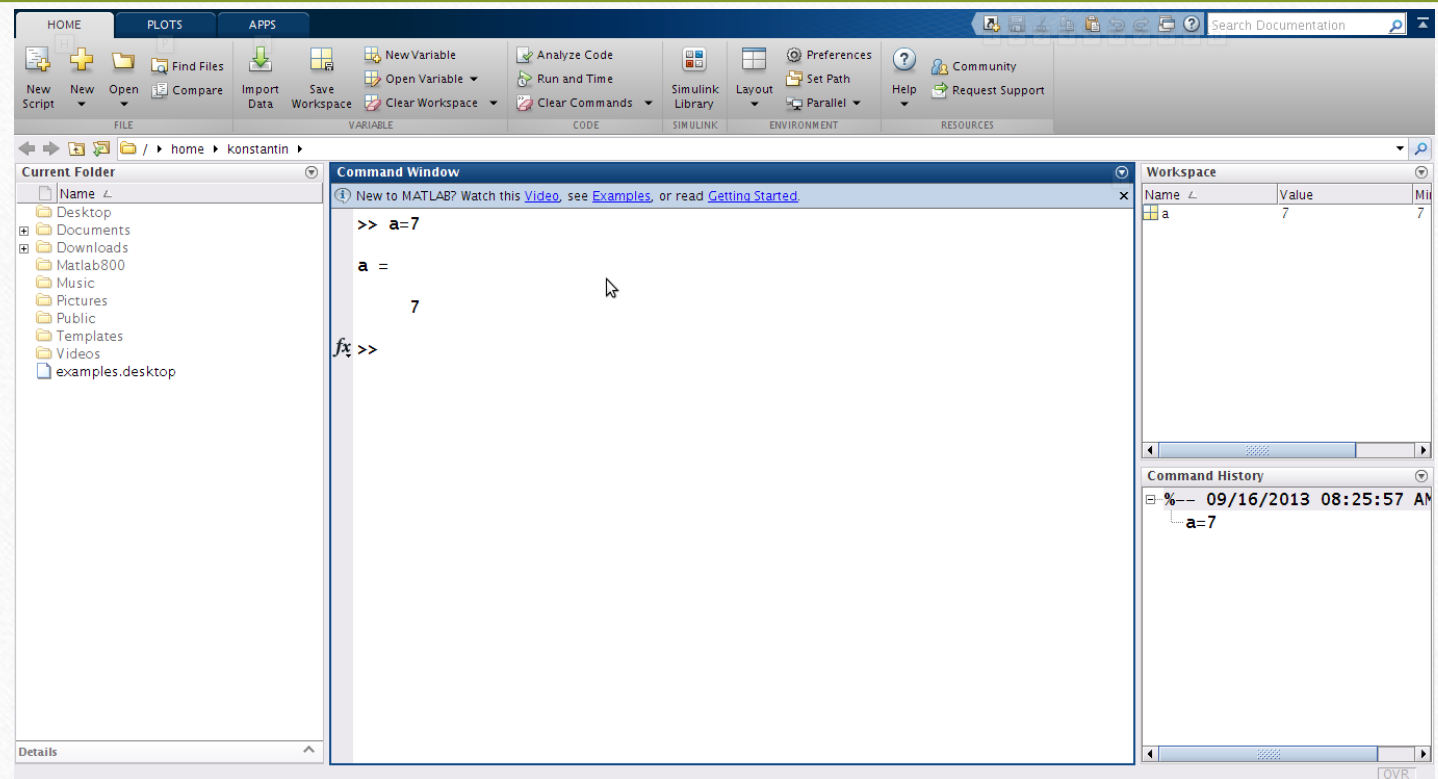
Introduction to Matlab

- MATLAB = MATrix LABoratory
- Rich sets of specialized functions—encapsulated in *toolboxes*
- MATLAB documentation:

Command line help

<http://www.mathworks.com>

MATLAB R2013b Environment



MATLAB Data Types

- MATLAB's basic data type is the *matrix* (or *array*).
- All data are considered to be matrices
- Single values are considered by MATLAB to be 1×1 matrices.

MATLAB Data Class

- **uint8** 8-Bit unsigned integers
- **uint16** 16-Bit unsigned integers
- **uint32** 32-Bit unsigned integers
- **int8** 8-Bit signed integers
- **int16** 16-Bit signed integers
- **int32** 32-Bit signed integers
- **single** Single-precision floating numbers (4 bytes)
- **double** Double-precision floating numbers (8 bytes)
- **logical** Values are 0 (false) or 1 (true) (1 bytes)
- **char** Characters (2 bytes)

Command Window

New to MATLAB? Watch this [Video](#), see [Examples](#), or read [Getting Started](#).

```
>> x=5  
  
x =  
  
    5  
  
>> y=uint8(12)  
  
y =  
  
    12  
  
>> whos x  
Name      Size      Bytes    Class    Attributes  
x         1x1         8    double  
  
>> whos y  
Name      Size      Bytes    Class    Attributes  
y         1x1         1    uint8  
  
>> class(x)  
  
ans =  
  
double  
fx >> |
```

OVR

class(), whos

```
Command Window
New to MATLAB? Watch this Video, see Examples, or read Getting Started.

>> 1/0
ans =
    Inf
>> Inf/Inf
ans =
    NaN
>> pi
ans =
    3.1416
>> i
ans =
    0 + 1.0000i
>> j
ans =
    0 + 1.0000i
fx >> |
```

Some Constants
and Variables

Inf
NaN
pi
ans
eps
i
j

```
Command Window
New to MATLAB? Watch this Video, see Examples, or read Getting Started.

>> pi

ans =

    3.1416

>> format compact
>> pi
ans =
    3.1416
>> format long
>> pi
ans =
    3.141592653589793
>> format rat
>> pi
ans =
    355/113
>> format bank
>> pi
ans =
    3.14
>> format hex
>> pi
ans =
    400921fb54442d18
fx >> |
```

Display format
compact/loose
long/short
longE
rat
hex
bank
...

```
>> st = 'hello'
```

```
st =  
hello
```

```
>> class(st)
```

```
ans =  
char
```

```
>> whos st
```

Name	Size	Bytes	Class	Attributes
st	1x5	10	char	

Strings are 1xn arrays of char

```
>> greeting = [st, ' world']
```

```
greeting =  
hello world
```

```
>> [greeting, 'it is', num2str(10), ' o'clock']
```

```
ans =  
hello worldit is10 o'clock
```

```
>> strcmpi('ABC','abc')
```

```
ans =  
1
```

```
>> strcmp('ABC','abc')
```

```
ans =  
0
```

fx >>

Structures

```
Command Window
>> student.name = 'bob'

student =

    name: 'bob'

>> student.age = 20

student =

    name: 'bob'
    age: 20

>> whos student
```

Name	Size	Bytes	Class	Attributes
student	1x1	366	struct	

fx >> |

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Cell Arrays

```
Command Window
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>> address = {'kaskelen','SDU'}
address =
    'kaskelen'    'SDU'
>> student = {'vasya', 20, address}
student =
    'vasya'    [20]    {1x2 cell}
>> student(1) %this way you actually get a "wrapper" over the 1st element
ans =
    'vasya'
>> whos ans
  Name      Size      Bytes  Class  Attributes
  ans       1x1         122   cell
>> student{1} % the actual variable
ans =
vasya
>> whos ans
  Name      Size      Bytes  Class  Attributes
  ans       1x5         10   char
>> whos student
  Name      Size      Bytes  Class  Attributes
  student   1x3         600   cell
fx >>
```

```
>> s = @sin
s =
    @sin
>> s(pi/2)
ans =
     1
>> myfunc = @(x,y) x+y
myfunc =
    @(x,y)x+y
>> myfunc(5,6)
ans =
    11
>> superfunction = @(x,f) f(x)
superfunction =
    @(x,f)f(x)
>> superfunction(pi/2, sin)
Error using sin
Not enough input arguments.
>> superfunction(pi/2, @sin)
ans =
     1
```

fx >> |

Function Handles
similar to
function pointers in C/C++
delegates in C#

Command Window

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```
>> A = input('Enter a matrix: ');  
Enter a matrix: [1 2 3; 4 5 6; 7 8 9]
```

```
>> disp(A)
```

```
    1    2    3  
    4    5    6  
    7    8    9
```

input, disp

```
>> text = input('Enter a text: ','s');  
Enter a text: i like to move it, move it ...
```

```
>> disp(text)
```

```
i like to move it, move it ...
```

fx

```
>> |
```

Command Window

```
>> A=magic(5)
A =
    17    24     1     8    15
    23     5     7    14    16
     4     6    13    20    22
    10    12    19    21     3
    11    18    25     2     9

>> A(2,3)
ans =
     7

>> A(4)
ans =
    10

>> A(2:3,1:2)
ans =
    23     5
     4     6

>> A(1:end,1:2)
ans =
    17    24
    23     5
     4     6
    10    12
    11    18

>> A(2,:)
ans =
    23     5     7    14    16

fx >> |
```

Array and Matrix
Indexing and Slicing

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Standard Arrays

- **zeros(m,n)** creates an $m \times n$ matrix of zeros.
- **ones(m,n)** creates an $m \times n$ matrix of ones.
- **true(m,n)** creates an $m \times n$ matrix of logical ones.
- **false(m,n)** creates an $m \times n$ matrix of logical zeros.
- **eye(n)** returns an $n \times n$ identity matrix.
- **magic(m)** returns a magic square2 of order m.
- **rand(m,n)** creates an $m \times n$ matrix whose entries are pseudorandom numbers uniformly distributed in the interval $[0, 1]$.
- **randn(m,n)** creates an $m \times n$ matrix whose entries are pseudorandom numbers that follow a normal (i.e., Gaussian) distribution with mean 0 and variance 1.

```
>> A = zeros(3,5)
```

```
A =  
    0    0    0    0    0  
    0    0    0    0    0  
    0    0    0    0    0
```

```
>> A = ones(3)
```

```
A =  
    1    1    1  
    1    1    1  
    1    1    1
```

```
>> A = ones(3)*3
```

```
A =  
    3    3    3  
    3    3    3  
    3    3    3
```

```
>> A = eye(3)
```

```
A =  
    1    0    0  
    0    1    0  
    0    0    1
```

```
>> A = true(3)
```

```
A =  
    1    1    1  
    1    1    1  
    1    1    1
```

```
>> whos ans
```

Name	Size	Bytes	Class	Attributes
ans	1x5	40	double	

```
>> rand(1,5)
```

```
ans =  
    0.8147    0.9058    0.1270    0.9134    0.6324
```

```
fx >> |
```

zeros, ones, eye, true, rand ...

File Edit Text Go Cell Tools Debug Parallel Desktop Window Help

Current Folder: /home/konst

Shortcuts How to Add What's New

Editor - /home/konst/myscript.m

```
1 - A=1:5;  
2 - B=5:-1:1;  
3 - disp(A);  
4 - disp(B);  
5 - disp(A+B);
```

Command Window

```
>> myscript  
1      2      3      4      5  
5      4      3      2      1  
6      6      6      6      6  
  
fx >>
```

M-file (script)

script Ln 5 Col 11 OVR

Editor - /home/konst/myfunc.m

```
1 function [sum pro] = myfunc(X,Y)
2 % returns sum and product of X and Y
3 sum = X+Y;
4 pro = X.*Y;
```

M-file (function)

```
>> X=1:5
```

```
X =
```

```
1 2 3 4 5
```

```
>> Y=5:9
```

```
Y =
```

```
5 6 7 8 9
```

```
>> help myfunc
```

```
returns sum and product of X and Y
```

```
>> [s,p]=myfunc(X,Y)
```

```
s =
```

```
6 8 10 12 14
```

```
p =
```

```
5 12 21 32 45
```

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FILE EDIT NAVIGATE BREAKPOINTS RUN

Current Folder: / > home > konstantin >

Editor - ifscript.m

```
1 - mark = input('Enter mark:');
2
3 - if mark<50
4 -     disp('fail')
5 - elseif mark<75
6 -     disp('no stipend')
7 - else
8 -     disp('zadr')
9 - end
```

Command Window

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```
>> ifscript
Enter mark:60
no stipend
>> ifscript
Enter mark:49
fail
>> ifscript
Enter mark:80
zadr
fx >>
```

if ... [elseif ...] [else ...] end

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Editor - forloop.m

```
1 - sum = 0;  
2 - for i=1:100  
3 -     sum = sum + i;  
4 - end  
5 - disp(sum);  
6
```

Command Window

```
>> type forloop  
  
sum = 0;  
for i=1:100  
    sum = sum + i;  
end  
disp(sum);  
  
>> forloop  
5050  
>> type whileloop  
  
i=1;  
sum = 0;  
while i<100  
    sum = sum + i;  
    i = i + 1;  
end  
  
disp(sum);  
>> whileloop  
4950  
fx >>
```

for/while ... end

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Operator	Name	Arithmetic Operations	MATLAB Function
+	Array and matrix addition		<code>plus(a,b)</code>
-	Array and matrix subtraction		<code>minus(a,b)</code>
.*	Element-by-element array multiplication		<code>times(a,b)</code>
*	Matrix multiplication		<code>mtimes(a,b)</code>
./	Array right division		<code>rdivide(a,b)</code>
.\	Array left division		<code>ldivide(a,b)</code>
/	Matrix right division		<code>mrdivide(a,b)</code>
\	Matrix left division		<code>mldivide(a,b)</code>
.^	Array power		<code>power(a,b)</code>
^	Matrix power		<code>mpower(a,b)</code>
.'	Vector and matrix transpose		<code>transpose(a)</code>
'	Vector and matrix complex conjugate transpose		<code>ctranspose(a)</code>
+	Unary plus		<code>uplus(a)</code>
-	Unary minus		<code>uminus(a)</code>
:	Colon		<code>colon(a,b)</code> or <code>colon(a,b,c)</code>

Exercises:

1. Perform matrix multiplication on two matrices.

$$\mathbf{X} = [1 \ 2 \ -2; \ 0 \ -3 \ 4; \ 7 \ 3 \ 0]$$

$$\mathbf{Y} = [1 \ 0 \ -1; \ 2 \ 3 \ -5; \ 1 \ 3 \ 5]$$

$$\mathbf{X} * \mathbf{Y}$$

2. Perform element-by-element multiplication.

$$\mathbf{X} .* \mathbf{Y}$$

3. Perform another matrix multiplication on two matrices.

$$\mathbf{X} = \text{eye}(3,4)$$

$$\mathbf{Y} = \text{rand}(4,2)$$

$$\mathbf{X} * \mathbf{Y}$$

$$\mathbf{Y} * \mathbf{X}$$

Specialized Matrix Operations

Name	MATLAB Operator or Function
Matrix transpose	Apostrophe (') operator
Inversion	inv function
Matrix determinant	det function
Flip up and down	flipud function
Flip left and right	fliplr function
Matrix rotation	rot90 function
Matrix reshape	reshape function
Sum of the diagonal elements	trace function

Operator

Name

<

Less than

<=

Less than or equal to

>

Greater than

>=

Greater than or equal to

==

Equal to

~=

Not equal to

Relational Operators

```
Command Window
>> A=1:3
A =
     1     2     3
>> B=3:-1:1
B =
     3     2     1
>> A>B
ans =
     0     0     1
>> A==B
ans =
     0     1     0
>> whos ans
Name      Size      Bytes  Class      Attributes
ans       1x3          3   logical
```

fx >> |

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Logical Operators and Logical Functions

Operator	Name
&	AND
	OR
~	NOT

Function	Description
xor	Performs the exclusive-or (X OR) between two operands
all	Returns a 1 if all the elements in a vector are nonzero or a 0 otherwise. Operates columnwise on matrices
any	Returns a 1 if any of the elements in a vector are nonzero or a 0 otherwise. Operates columnwise on matrices

Logical Operators and Logical Functions

```
>> A = logical(round(rand(1,5)))
A =
     0     1     1     0     0
>> B = logical(round(rand(1,5)))
B =
     0     1     0     1     0
>> whos A B
  Name      Size      Bytes  Class  Attributes
  ----      -
  A         1x5         5  logical
  B         1x5         5  logical

>> A&B
ans =
     0     1     0     0     0
>> A|B
ans =
     0     1     1     1     0
>> ~A
ans =
     1     0     0     1     1
>> and(A,B)
ans =
     0     1     0     0     0
>> xor(A,B)
ans =
     0     0     1     1     0
>> all(A)
ans =
     0
>> any(A)
ans =
     1
fx >> |
```

```
>> why
The programmer suggested it.
>> why
Barney suggested it.
>> why
To please a very terrified and smart and tall engineer.
>> why
The tall system manager obeyed some engineer.
>> why
To satisfy some programmer.
>> why
Damian wanted it that way.
>> why
Can you rephrase that?
>> why
Because Damian wanted it that way.
>> why
How should I know?
>> why
Because they asked the terrified and smart and tall and tall programmer.
>> why
To fool a young tall hamster.
>> why
For the love of a bald and terrified mathematician.
>> why
It's your karma.
>> why
Some terrified and rich system manager knew it was a good idea.
>> why
Don't you have something better to do?
>> why
He suggested it.
>> why
A terrified and good and not very rich engineer helped the bald programmer.
fx >> |
```

Fun in MATLAB

Homework

- Exercises in the end of **Chapter 3 of textbook.**